ASTANA IT UNIVERSITY

«Approved»
by the Dean
__________Syzdykova Z.A.
«___» of __________ 2024

Syllabus
Academic Year 2024-2025

1. General Information	
Course title	**Numerical Methods**
Degree cycle (level)/ major	Cyber Security
Year, term	2, 1
Number of credits	5
Language of delivery:	English
Prerequisites	Calculus
Postrequisites	**Numerical Methods for Differential Equations**
Lecturer(s)	Rysbaiuly Bolatbek, DPh&MS, Professor in Mathematical and Computer Modeling, rysbaiuly.b@gmail.com Astana IT University, Expo. C1 block, 3rd floor, office C1.1.332
2. Goals, Objectives and Learning Outcomes of the Course	
1. Course Description	Numerical Methods is a core course at Astana IT University; it plays an important role in understanding science, technology, economics and computer science, as well as other disciplines. This course, Numerical Methods , covers difference derivatives, algorithms and methods for solving the Cauchy problem for differential equations, interpolation problems, approximate methods for calculating definite integrals, approximate methods for solving systems of linear algebraic equations, solving nonlinear and systems of nonlinear algebraic equations.
2. Course Goal(s)	The objective of the course is that upon completion of the course, students will be able to use computational methods for solving SLAE and NSLAE. Recognize and master from a user's perspective the concepts and basic computational methods of algebra and analysis. Course Objectives: • Introduce the basic ideas of numerical methods of algebra and analysis, including computational methods for solving SLAE and NSLAE. • Apply the basic concepts of numerical methods of algebra and analysis to various applications. Course Learning Outcomes Students who successfully complete the course will be able to: • Solve real-world algebra and analysis processes numerically (using a computer), . • Create algorithms and codes for solving real-world algebra and analysis processes.
3. Course Objectives:	Students should learn the following concepts: • Solution of a system of linear algebraic equations (SLAE). Method, algorithm and program. • Solution of a system of nonlinear algebraic equations (SNAE).

		Method, algorithm and program. • Splines of the first, second and third order. Convergence, algorithm and program. • Numerical methods for calculating definite integrals. Convergence, algorithm and program. • Numerical methods for solving the Cauchy problem for an ordinary differential equation (ODE). Convergence, algorithm and program.
4.	Skills & Competences	Upon completion of this course, students should demonstrate competence in the following skills in solving algebra and analysis problems: • Method of checking the convergence and stability of numerical methods for solving SLAE problems; • Method of checking the convergence of numerical methods for solving SNAE problems; • Method of checking the stability of numerical methods for solving interpolation problems; • Approximate method for calculating definite integrals. Checking the convergence of approximate methods.
5.	Course Learning Outcomes:	By the end of this course, students will be able to understand concepts related to algebra and analysis problems, namely, numerical methods for solving SLAE, SNAE, interpolation problems, approximate methods for solving the Cauchy problem for ODE, approximate calculations of definite integrals. The student should be able to establish the convergence and stability of the studied approximate methods.
6.	Methods of Assessment	• Homework • Quiz • Midterm • Pre-Final and Final Exams
7.	Reading List	Assigned reading materials and presentations should be read prior to class. Class lectures and discussions will proceed with supplemental and advanced topics, which could be difficult to understand unless students have read the assigned material. Readings are listed in the schedule section. All necessary updates and/or changes to the course will be reflected in the Learning Management System (moodle.astanait.edu.kz). Basic Literature: **1.** Tai-Ran-HSU.Applied Engineering Analysis. San Jose, USA, 2018 **2.** Jeffrey R. Chasnov. Numerical Methods for Engineers, 2012, Hong Kong **3.** Ahmed Ayash. Applied Engineering Analisys Course I, II. 2019, 391 p. Supplementary literature: **4.** А. А. Самарский, А.В. Гулин. Численные методы, 1989, 432c.

	5. Xin-She-Yang, Applied Engineering Mathematics, 2007, Cambridge UK254 p.
1. Resources	Online journals, articles, papers, books, and internet resources, as well as online emulators and online software for simulation.
2. **Course policy**	**Course and University policies include:** **Attendance:** Attendance is not allocated any grading points in the marking scheme, but is compulsory to pass the course. Normally, students are required to achieve course attendance of a minimum 70% to get admitted to the examination rubric. In case a student misses 30% or more class sessions without a valid excuse, the instructor has the right to mark him as "not graded." In such case, a student is not admitted to the exam and automatically fails the course. It should be NOTED that in cases when a student is excused for 30% of the scheduled class sessions or more, he or she has to study material provided under the course on their own. Course instructors might provide additional opportunities to submit missed graded pieces of work during office hours or conduct alternative assessment exercises using the method of his or her choosing. **Preparation for Class**: Class participation is a very important part of the learning process in this course. Although not explicitly graded, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties: - Offers a different and unique, but relevant, perspective; - Contributes to moving the discussion and analysis forward; - Builds on other comments. **Class work**: Each lecture and practical lesson lasts 50 minutes for offline classes and 40 minutes for online classes. Students are expected to complete all readings and assignments ahead of time, attend class regularly, and participate in class discussions. In case of systemic student misconduct, the student can be dispensed from the classes. **Being late on class:** When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, lateness can become chronic and spread throughout the class if left unchecked. Therefore, being late to class is not welcome and can be restriction activities by the course instructor. **Attestation I and II:** Students who score less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically fail the course. **Home work / Assignments**: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for

	the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In the case of using someone's work (papers, articles, and other publications), all works must be properly cited. Failure to cite work will result in cheating from the students and may be the subject of additional disciplinary measures. **Late submissions:** Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All grades are based on a percentage grading scale. *In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances of the instructor.* **Final exam** At the completion of this course, the Final Exam will be given in the written form **Laptops and mobile devices** can only be used for classroom purposes when directed by the course instructor. Misuse of laptops or handheld devices will be considered a breach of discipline, and the instructor will initiate appropriate action. **Online lessons** can be used in case there won't be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run online lessons is Microsoft Teams for video calls and live webinars and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround. **Cheating and plagiarism** are defined in the Academic conduct policies of the university and include: 1. Submitting work that is not your own papers, assignments, or exams; 2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation; 3. Submitting or using falsified data; 4. Submitting the same work for credit in two courses without prior consent of both instructors. Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work, and further actions will also be taken regarding the academic conduct policies of the university. **Academic Conduct Policies of the university:** The full texts of all the academic conduct codes will be posted to the students using the Learning Management System (moodle.astanait.edu.kz). **Contacting the Course instructor**: The easiest and most reliable way to get in touch with the course instructor is by email. Students must feel free to send an email if they have a question related to the course. The

	instructor responds as soon as they can but not always instantaneously. Besides that, students are also welcome to arrange a one-to-one meeting with the instructor during office hours to discuss the class offline and online.

* *Final exam can be given by the instructor in any other form such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.*

3. Course Content

#	Abbreviation	Meaning
1	ISIS	Instructor-supervised independent work
2	SIS	Students' independent work
3	IP	Individual project
4	PA	Practical assignment
5	LW	Laboratory work
6	MCQ	Multiple choice quiz

3.1 Lecture, Practical/Seminar/Laboratory Session Plans

Week No	Course topics	Lectures (h/w)	Practical sessions (h/w)	TSIS (h/w)	SIS (h/w)
1	Solution of SLAE. Cramer's method. Systems of linear algebraic equations depending on a parameter.	3	2	3	6
2	Case n=m Approximate methods for solving SLAE. Jacobi method. Convergence. Algorithm and program Approximate methods for solving SLAE. Seidel's method. Convergence. Algorithm and program	3	2	3	6
3	Approximate methods for solving SNAU. 1D case. Bisection method. Tangent method. Convergence.	3	2	3	6
4	Approximate methods for solving SNAU. 1D case. Newton's method. Modified Newton's method. Convergence. Algorithm. Program.	3	2	3	6
5	Approximate methods for solving SNAU. Case nD, n≥2. Newton's method. Modified Newton's method. Convergence. Algorithm. Program.	1	2	3	6
5	**Midterm exam**	2	0	0	0
6	Interpolation. Splines. 1D case. Calculation formulas. Convergence. Error. Algorithm.	3	2	3	6

7	Splines. 2D and 3D case. Calculation formulas. Convergence. Error. Algorithm.	3	2	3	6
8	Approximate calculation of a definite integral. Rectangle and trapezoid formula. Convergence. Error. Algorithm.	3	2	3	6
9	Approximate calculation of a definite integral. Simpson's formula. Convergence. Error. Algorithm. Difference derivatives. Error.	1	0	3	6
	Endterm exam	2	0		
10	Approximate methods for solving the Cauchy problem for ODE. Explicit and implicit calculation schemes. Algorithm of the Runge-Kutta method. Convergence. Algorithm.	3	2	3	6
	Pre-final exam				
	Total hours: 140	**30**	**20**	**30**	**60**

3.2 List of Assignments for Student Independent Study

№	Assignments (topics) for independent study	Hours	Recommended literature and other sources (links)	Form of submission
1	**2**	**3**	**4**	**5**
1	Solution of SLAE. Gauss method. Matrix method.	9	Reading books, [3-5]	
2	Case n=m Approximate methods for solving SLAE. Jacobi method. Seidel's method. Convergence. Algorithm and program	9	Reading books, [3-5]	Exercises
3	Approximate methods for solving SNAU. 1D case. Bisection method. Tangent method. Convergence.	9	Reading books, [3-5]	Exercises
4	Approximate methods for solving SNAU. 1D case. Newton's method. Modified Newton's method. Convergence. Algorithm. Program.	9	Reading books, [3-5]	
5	Approximate methods for solving SNAU. Case nD,$n \geq 2$. Newton's method. Modified Newton's method. Convergence. Algorithm. Program.	9	Reading books, [3-5]	Exercises
6	Interpolation. Splines. 1D case. Calculation formulas. Convergence.	9	Reading books, [3-5]	

	Error. Algorithm.			
7	Splines. 2D and 3D case. Calculation formulas. Convergence. Error. Algorithm.	9	Reading books, [3-5]	Exercises
8	Approximate calculation of a definite integral. Rectangle and trapezoid formula. Convergence. Error. Algorithm.	9	Reading books, [3-5]	
9	Approximate calculation of a definite integral. Simpson's formula. Convergence. Error. Algorithm. Difference derivatives. Error.	9	Reading books, [3-5]	Exercises
10	Approximate methods for solving the Cauchy problem for ODE. Explicit and implicit calculation schemes. Algorithm of the Runge-Kutta method. Convergence. Algorithm.	9	Reading books, [3-5]	Exercises
	Total hours	**90**		

4. Student Performance Evaluation System for the Course

Period	Assignments	Number of points	Total
1st attestation	**Assignments:** assignment 1, assignment 2, assignment 3, **Mid Term**	**60** 20 20 20 **40**	100
2nd attestation	**Assignments:** assignment 4, assignment 5, assignment 6, **End Term**	**60** 20 20 20 **40**	100
Final exam*	Written Exam		100
Total	**0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final**		**100**

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	Good
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	

C	2,0	65-69	Satisfactory
C-	1,67	60-64	
D+	1,33	55-59	
D	1,0	50-54	
FX	0	25-49	Fail
F	0	0-24	

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment
80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement
60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and material taught
25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught
0-24	No significant assessable material, absent or assessment missing a must pass component

5. Methodological Guidelines

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **ISIS (Instructor Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and end term exams is complex.
- **Final assessment** is a combination of both individual (team) project (report) and oral presentation (slides) to evaluate the students' academic performance and professional skills. At the completion of this course each student has to submit an online project version conforming to the project outline, as well as to prepare and present a slide presentation that follows the presentation outline. Project iterations would be required to submit as well. Students should submit the written Report of Final project and slide-Presentation 3 days before the Final exam day.

If necessary, the teacher has the right to change points 3.1 and 3.2.

6. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	Job title	Date	Sign
Rysbaiuly Bolatbek	Professor	02/09/2024	

Director **Sergaziyev M.Zh**